



Team Name: ASTRATEK
Project Name: IMPACTOR 2025

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Benefits

- 1 Simulating complex celestial phenomena—such as asteroid trajectories or mitigation attempts—lets researchers experiment safely and iteratively without costly physical trials.
- 2 Virtual test beds enable rapid hypothesis testing, offering insights that may be impractical or dangerous to reproduce in real life.
- 3 Simulation accelerates learning and collaboration by making dynamic scenarios observable from multiple perspectives.

Weaknesses & Challenges

- 1 Obtaining, cleaning, and validating high-quality astrophysical datasets is time-consuming and may introduce bias or uncertainty.
- 2 Translating physical mechanics—gravity, material behavior, collision energy—into accurate numerical models requires intensive calibration and domain expertise.
- 3 Balancing fidelity and computational efficiency is difficult, especially when simulating long-term orbital evolution.
- 4 Communicating the limitations of simulated results to non-technical audiences remains an ongoing challenge.

Solution Overview

- 1 Combine astronomical datasets with machine learning pipelines to identify potentially hazardous asteroids (PHAs) before they become imminent threats.
- 2 Integrate predictive models with a physics-based simulation engine that evaluates various mitigation strategies (deflection missions, kinetic impactors, gravitational tractors, etc.).
- 3 Present the results in an interactive environment that shows orbital dynamics, collision probabilities, and mitigation outcomes.

Target Users

- 1 Enthusiasts, educators, students, policy analysts, and anyone curious about planetary defense—even without deep technical knowledge.
- 2 Organizations or citizen-science groups seeking accessible tools to visualize asteroid risks and explore mitigation concepts.

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Expected Results

- 1 A dynamic solar-system simulation with accurate orbital data for planets, moons, and tracked asteroids.
- 2 Automated prediction of potential asteroid collisions based on machine learning analysis of orbital changes.
- 3 Scenario-based mitigation simulations, illustrating how different strategies alter collision outcomes.
- 4 Clear visualizations that help users understand risk levels and compare proposed interventions.

Main Features

- 1 Orbit Visualization – 3D trajectories and timeline controls for celestial bodies and asteroids.
- 2 Asteroid Tracking – Real-time updates from cataloged objects, with alerts for changes in risk classification.
- 3 PHA Detection – Machine learning classifiers that score asteroids by potential hazard level.
- 4 Collision Forcing – Sandbox mode to model hypothetical impact scenarios for research or education.
- 5 Mitigation Simulation – Tools to test deflection missions, energy outputs, and post-mitigation orbital shifts.
- 6 Analytics & Reporting – Summaries of risk, mission performance, and recommended next steps.

Summary

This project delivers an accessible planetary-defense sandbox: it forecasts asteroid collisions, explores mitigation tactics, and communicates insights to experts and newcomers alike. By uniting machine learning with orbital simulations, the platform reveals both the promise and the complexity of safeguarding Earth from celestial threats.